

LONS-LE-SAUNIER, France

WMO: 073900

Lat: 46.6928N

Lon: 5.5177E

Elev: 279

StdP: 98.02

Time Zone: 1.00 (EUC)

Period: 99-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-5.3	-10.3	1.6	-5.1	-8.1	2.0	-4.2	11.4	8.5	10.1	7.5	1.4	100	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	31.5	20.4	29.5	20.1	27.8	19.7	21.8	28.5	20.9	27.7	20.2	26.4	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.7	25.5	18.6	13.9	24.2	17.8	13.2	23.1	64.9	28.8	61.6	27.9	59.0	26.7	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.0	6.4	-9.3	33.7	2.6	2.8	-11.2	35.7	-12.7	37.4	-14.1	38.9	-16.0	41.0		
			-9.8	23.3	2.4	0.9	-11.5	24.0	-13.0	24.5	-14.4	25.0	-16.1	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	2.7	4.5	7.7	11.2	15.6	19.1	20.3	19.6	16.8	12.9	6.8	3.2	(6)
		DBStd	7.47	4.59	4.14	4.22	3.64	3.54	4.02	3.53	3.49	3.56	3.82	4.28	4.61	(7)
		HDD10.0	857	229	158	94	28	0	0	0	1	17	113	214		(8)
		HDD18.3	2656	486	388	330	213	98	41	19	23	70	172	347	469	(9)
		CDD10.0	1487	2	3	23	66	176	272	318	298	204	106	16	3	(10)
		CDD18.3	246	0	0	0	1	14	63	79	63	23	3	0	0	(11)
		CDH23.3	2052	0	0	0	11	115	552	691	543	133	7	0	0	(12)
		CDH26.7	633	0	0	0	0	18	161	233	204	17	0	0	0	(13)
(14)	Wind	WSAvg	2.6	2.6	2.9	3.1	3.1	2.7	2.5	2.2	2.1	2.4	2.6	2.5	2.8	(14)
(15)	Precipitation	PrecAvg	1380	112	101	97	112	121	110	107	106	110	139	151	125	(15)
(16)		PrecMax	1764	232	238	339	245	207	210	249	245	252	254	425	227	(16)
(17)		PrecMin	1070	46	22	17	21	42	27	45	16	30	42	35	4	(17)
(18)		PrecStd	187	50	54	66	61	48	49	48	55	60	59	83	53	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	13.8	15.5	20.6	25.0	28.9	33.0	34.1	37.7	28.7	24.3	18.5	15.4		(19)
(20)		0.4% MCWB	10.1	9.7	13.1	15.8	19.3	20.1	19.8	19.8	18.9	17.7	13.6	11.3		(20)
(21)		2% DB	12.2	13.7	17.9	22.0	26.2	30.1	31.2	31.6	26.8	21.7	16.5	12.9		(21)
(22)		2% MCWB	9.6	9.0	11.2	14.4	17.9	20.6	20.8	19.6	19.2	16.3	12.2	9.7		(22)
(23)		5% DB	11.1	11.9	15.7	20.0	24.3	28.4	29.3	28.8	24.9	20.2	14.4	11.2		(23)
(24)		5% MCWB	8.7	8.2	10.2	13.1	16.9	19.9	20.2	19.8	18.2	15.6	11.3	9.0		(24)
(25)		10% DB	9.5	10.5	14.1	17.7	22.4	26.7	27.3	26.3	22.8	18.6	12.8	9.8		(25)
(26)		10% MCWB	7.5	7.6	9.8	12.1	16.0	19.2	19.7	19.1	17.3	15.0	10.5	7.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	11.2	10.5	13.8	16.5	20.5	22.8	23.0	22.0	21.5	18.5	14.4	11.7		(27)
(28)		0.4% MCDB	12.6	13.6	20.1	23.1	26.7	29.5	29.8	28.6	26.7	22.4	17.2	13.9		(28)
(29)		2% WB	9.9	9.5	12.2	14.9	18.8	21.8	21.8	21.1	20.2	17.2	12.7	10.2		(29)
(30)		2% MCDB	12.0	12.5	15.9	20.7	25.1	28.2	29.0	28.7	25.0	20.3	15.3	12.2		(30)
(31)		5% WB	8.8	8.6	11.2	13.7	17.7	20.7	20.8	20.2	19.0	16.1	11.6	9.2		(31)
(32)		5% MCDB	10.7	11.4	14.6	19.1	23.0	27.0	27.6	27.3	23.4	19.6	14.2	11.2		(32)
(33)		10% WB	7.6	7.8	10.2	12.5	16.5	19.6	19.9	19.5	17.7	15.1	10.6	8.0		(33)
(34)		10% MCDB	9.3	10.3	13.7	17.0	21.2	25.3	26.1	25.6	21.7	18.2	12.7	9.5		(34)
(35)	Mean Daily Temperature Range	MDBR	5.5	6.6	8.2	9.2	9.4	10.3	10.0	9.7	9.3	7.4	5.9	4.9		(35)
(36)		5% DB MCDBR	6.3	8.4	11.0	12.9	12.3	12.4	12.7	13.4	11.2	9.1	7.7	6.1		(36)
(37)		5% WB MCWBR	4.4	5.4	6.3	7.0	6.4	5.7	5.5	5.8	5.4	5.2	4.8	4.3		(37)
(38)		5% WB MCDBR	5.9	7.1	9.0	11.9	11.0	11.7	12.0	11.8	10.1	8.5	7.0	5.9		(38)
(39)		5% WB MCWBR	4.3	4.8	5.4	6.7	6.1	5.6	5.5	5.7	5.5	5.3	4.9	4.5		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.329	0.361	0.396	0.402	0.407	0.404	0.398	0.377	0.362	0.328	0.306		(40)
(41)		taud	2.447	2.389	2.335	2.250	2.280	2.303	2.308	2.339	2.378	2.411	2.454	2.457		(41)
(42)		Ebn at Noon	787	838	860	858	864	859	856	847	833	788	752	745		(42)
(43)		Edh at Noon	68	88	107	128	129	126	124	116	102	85	68	61		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.12	1.95	3.24	4.46	5.21	6.13	5.88	5.09	3.92	2.42	1.29	0.92		(44)
(45)		RadStd	0.14	0.31	0.43	0.62	0.54	0.56	0.51	0.42	0.31	0.20	0.12	0.15		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page